

Network Traffic Analysis Using Wireshark on macOS

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Project Objective

The objective of this project was to capture and analyze network traffic generated by normal web browsing activity on a macOS device. Wireshark was used to inspect DNS requests, TCP connection establishment, TLS-encrypted traffic, network endpoints, conversations, and traffic volume

Lab Environment

Item	Details
Operating System	macOS
Packet Analysis Tool	Wireshark
Capture Interface	[Wi-Fi/en0]
Capture Duration	5 minutes
Browsing Activity	Google, Youtube, and Wikipedia
Capture File	network_capture.pcapng

Methodology

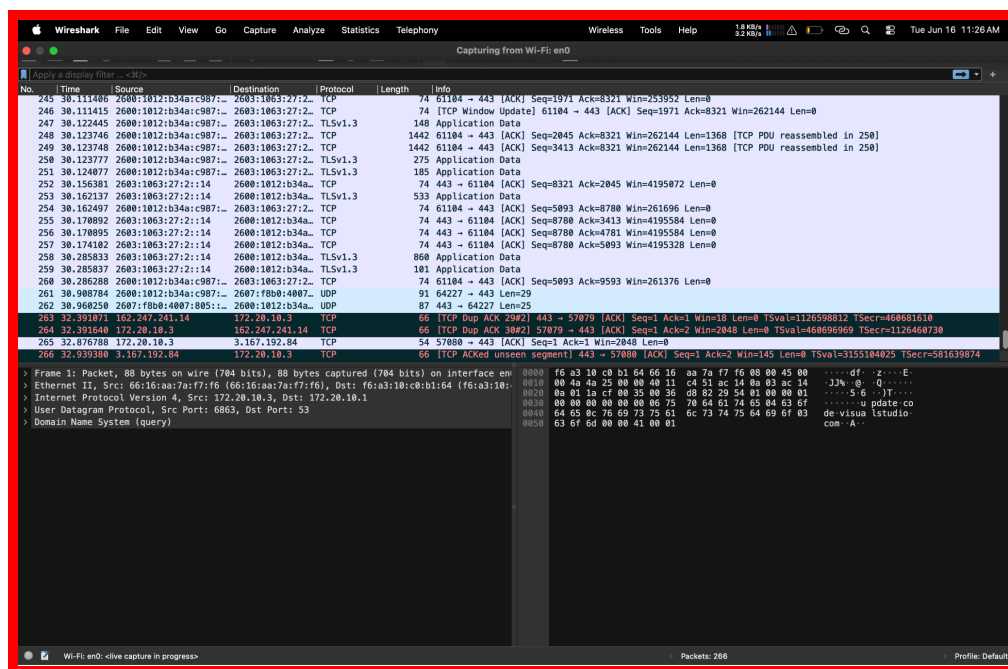
1. Wireshark was opened on the active Wi-Fi network interface
2. A live packet capture was started
3. Google, YouTube, and Wikipedia were visited during the capture
4. The capture was stopped after approximately 5 minutes
5. Wireshark display filters and Statistics tools were used to analyze DNS, TCP, TLS, endpoints, conversations, and traffic volume
6. The capture was saved as network_capture.pcapng

Findings and Analysis

Live Packet Capture

A live packet capture was conducted while browsing Google, YouTube, and Wikipedia. The packet list displayed communication between the local device and external servers. Multiple protocols were observed, including DNS, TCP, UDP, and TLS

Figure 1. Live packet capture of network activity generated during web browsing sessions involving Google, YouTube, and Wikipedia.



DNS Analysis

The DNS display filter was applied to identify domain name resolution activity. DNS queries were observed while the browser attempted to locate the IP addresses associated with websites visited during the capture.

DNS is important because users access websites by domain name, while network devices communicate using IP addresses. DNS translates a domain name, such as [google.com](https://www.google.com), into an IP address that the browser can use to establish a connection.

The capture showed DNS requests for domains associated with web browsing activity, including [google.com](https://www.google.com) and [wikimedia.org](https://www.wikimedia.org)

Figure 2. DNS traffic showing domain name resolution requests generated during web browsing

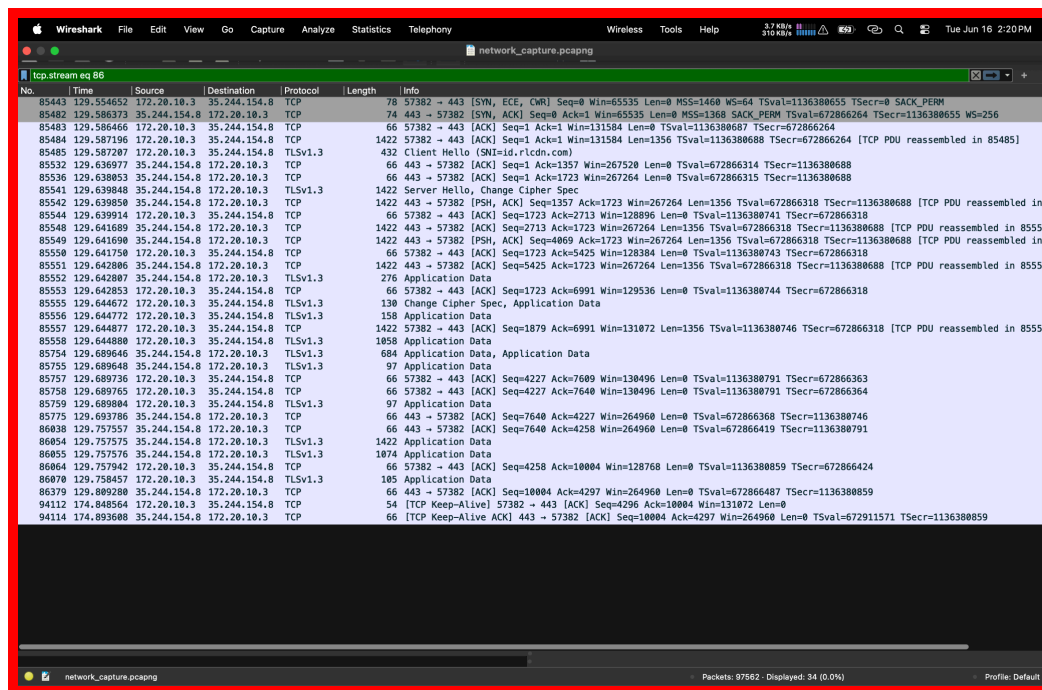
No.	Time	Source	Destination	Protocol	Length	Info
94.	173.773.	172.28.10.1	172.28.10.3	DNS	133	Standard query response 0x0000 AAAA en.wikipedia.org CNAME dyna...
94.	173.773.	172.28.10.1	172.28.10.3	DNS	121	Standard query response 0x3681 A en.wikipedia.org CNAME dyna...
94.	175.013.	172.28.10.3	172.28.10.1	DNS	88	Standard query 0x26db HTTPS update.code.visualstudio.com
94.	175.014.	172.28.10.3	172.28.10.1	DNS	88	Standard query 0x7543 AAAA update.code.visualstudio.com
94.	175.014.	172.28.10.3	172.28.10.1	DNS	88	Standard query 0x5ed7 A update.code.visualstudio.com
94.	175.059.	172.28.10.1	172.28.10.3	DNS	274	Standard query response 0x5ed7 A update.code.visualstudio.com
94.	175.068.	172.28.10.1	172.28.10.3	DNS	298	Standard query response 0x7543 AAAA update.code.visualstudio.c...
94.	175.182.	172.28.10.1	172.28.10.3	DNS	299	Standard query response 0x26db HTTPS update.code.visualstudio...
94.	176.271.	172.28.10.3	172.28.10.1	DNS	75	Standard query 0xbf20 HTTPS bam.nr-data.net
94.	176.271.	172.28.10.3	172.28.10.1	DNS	75	Standard query 0xd02c AAAA bam.nr-data.net
94.	176.272.	172.28.10.3	172.28.10.1	DNS	75	Standard query 0x1220 A bam.nr-data.net
94.	176.285.	172.28.10.1	172.28.10.3	DNS	159	Standard query response 0x1220 A bam.nr-data.net CNAME bam.ce...
94.	176.331.	172.28.10.1	172.28.10.3	DNS	201	Standard query response 0xd02c AAAA bam.nr-data.net CNAME bam...
94.	176.337.	172.28.10.1	172.28.10.3	DNS	201	Standard query response 0xbf20 HTTPS bam.nr-data.net CNAME bam...
94.	176.708.	172.28.10.3	172.28.10.1	DNS	81	Standard query 0xaf8c HTTPS commons.wikimedia.org
94.	176.708.	172.28.10.3	172.28.10.1	DNS	81	Standard query 0x915c AAAA commons.wikimedia.org
94.	176.708.	172.28.10.3	172.28.10.1	DNS	81	Standard query 0x0274 A commons.wikimedia.org
94.	176.758.	172.28.10.1	172.28.10.3	DNS	151	Standard query response 0xaf8c HTTPS commons.wikimedia.org CNA...
94.	176.759.	172.28.10.1	172.28.10.3	DNS	128	Standard query response 0x915c AAAA commons.wikimedia.org CNA...
94.	176.759.	172.28.10.1	172.28.10.3	DNS	116	Standard query response 0xb074 A commons.wikimedia.org CNAME d...
94.	177.011.	172.28.10.3	172.28.10.1	DNS	80	Standard query 0xb0b6 HTTPS upload.wikimedia.org
94.	177.012.	172.28.10.3	172.28.10.1	DNS	80	Standard query 0xd266 AAAA upload.wikimedia.org
94.	177.012.	172.28.10.3	172.28.10.1	DNS	80	Standard query 0x3df1 A upload.wikimedia.org
94.	177.012.	172.28.10.3	172.28.10.1	DNS	78	Standard query 0x5c83 HTTPS meta.wikimedia.org
94.	177.012.	172.28.10.3	172.28.10.1	DNS	78	Standard query 0x26ac AAAA meta.wikimedia.org
94.	177.013.	172.28.10.3	172.28.10.1	DNS	78	Standard query 0x5aff A meta.wikimedia.org
94.	177.016.	172.28.10.1	172.28.10.3	DNS	131	Standard query response 0xb0b6 HTTPS upload.wikimedia.org SOA...
94.	177.017.	172.28.10.1	172.28.10.3	DNS	188	Standard query response 0xd266 AAAA upload.wikimedia.org AAAA...
94.	177.022.	172.28.10.1	172.28.10.3	DNS	96	Standard query response 0x3df1 A upload.wikimedia.org A 190.35...
94.	177.058.	172.28.10.1	172.28.10.3	DNS	148	Standard query response 0x5c83 HTTPS meta.wikimedia.org CNAME...
94.	177.052.	172.28.10.1	172.28.10.3	DNS	125	Standard query response 0x26ac AAAA meta.wikimedia.org CNAME d...
94.	177.052.	172.28.10.1	172.28.10.3	DNS	113	Standard query response 0x5aff A meta.wikimedia.org CNAME dyna...
94.	177.258.	172.28.10.3	172.28.10.1	DNS	91	Standard query 0xf6b6 HTTPS content-autofill.googleapis.com
94.	177.258.	172.28.10.3	172.28.10.1	DNS	91	Standard query 0xb0e47 AAAA content-autofill.googleapis.com
94.	177.251.	172.28.10.3	172.28.10.1	DNS	91	Standard query 0x2be4 A content-autofill.googleapis.com
94.	177.257.	172.28.10.1	172.28.10.3	DNS	148	Standard query response 0xf6b6 HTTPS content-autofill.googleleap...
94.	177.257.	172.28.10.1	172.28.10.3	DNS	203	Standard query response 0xb0e47 AAAA content-autofill.googleleap...
94.	177.257.	172.28.10.1	172.28.10.3	DNS	153	Standard query response 0x2be4 A content-autofill.googleleap...
95.	177.713.	172.28.10.3	172.28.10.1	DNS	78	Standard query 0xec61 HTTPS auth.wikimedia.org
95.	177.713.	172.28.10.3	172.28.10.1	DNS	78	Standard query 0xf733 AAAA auth.wikimedia.org
95.	177.714.	172.28.10.3	172.28.10.1	DNS	78	Standard query 0xb4b3 A auth.wikimedia.org
95.	177.749.	172.28.10.1	172.28.10.3	DNS	148	Standard query response 0xec61 HTTPS auth.wikimedia.org CNAME...
95.	177.750.	172.28.10.1	172.28.10.3	DNS	125	Standard query response 0xf733 AAAA auth.wikimedia.org CNAME d...
95.	177.750.	172.28.10.1	172.28.10.3	DNS	113	Standard query response 0xb4b3 A auth.wikimedia.org CNAME dyna...

TCP Three-Way Handshake

A TCP connection was analyzed to observe the TCP three-way handshake. TCP is a connection-oriented protocol that establishes a reliable communication session before data is exchanged

Handshake	Meaning
SYN	The client initiated a request to establish a connection
SYN-ACK	The server acknowledged the request and responded with its own synchronization request.
ACK	The client acknowledged the server response, completing the connection setup

Figure 3. TCP three-way handshake establishing a reliable network connection between the client and remote web server prior to encrypted TLS communication.



The TLS display filter was used to inspect encrypted web traffic. The capture showed TLS messages including Client Hello, Server Hello, and encrypted Application Data. In some connections, a Change Cipher Spec message was also observed

The Client Hello and Server Hello messages are part of the TLS handshake. They allow the client and server to negotiate security settings before encrypted data is exchanged. After the TLS handshake, web traffic appears as Application Data because the contents are encrypted

[illegible]

Figure 4. TLS encrypted communication protecting web traffic from interception. Client Hello and Server Hello messages establish a secure session before encrypted application data is encrypted

Protocol Hierarchy Analysis

Wireshark's protocol hierarchy feature was used to identify the protocols present during the capture. The results showed the distribution of network traffic across protocols such as TCP, UDP, DNS, and TLS

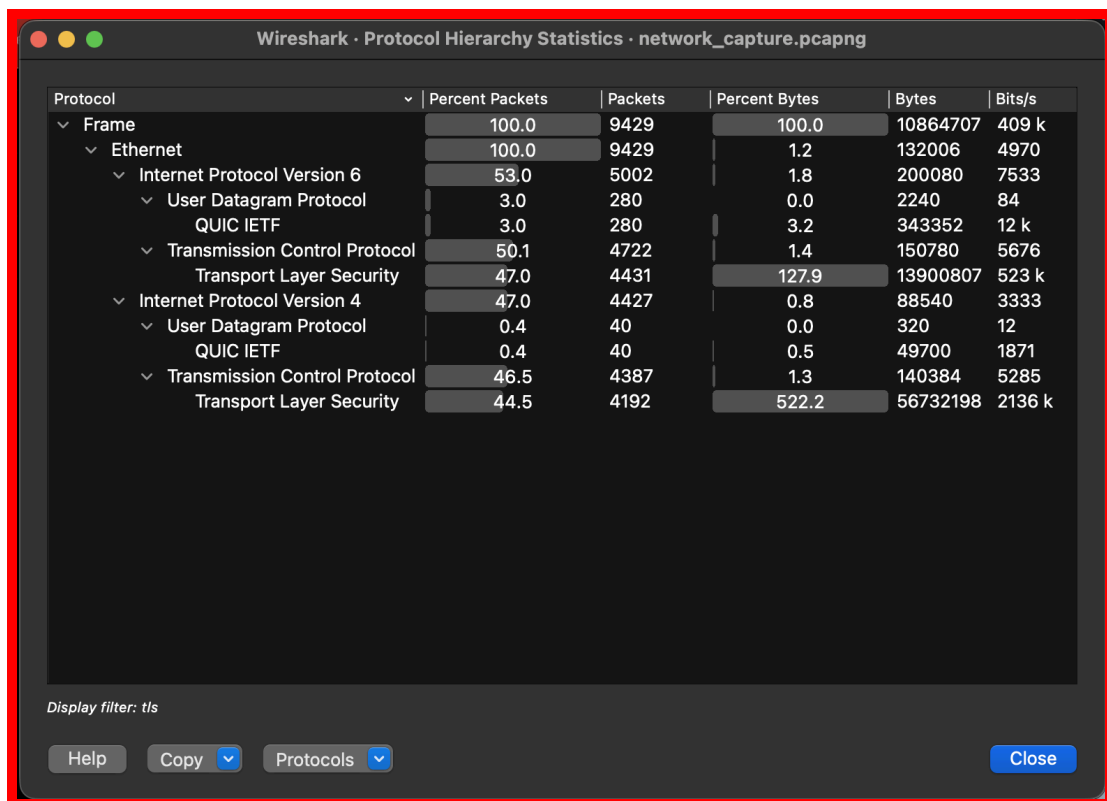


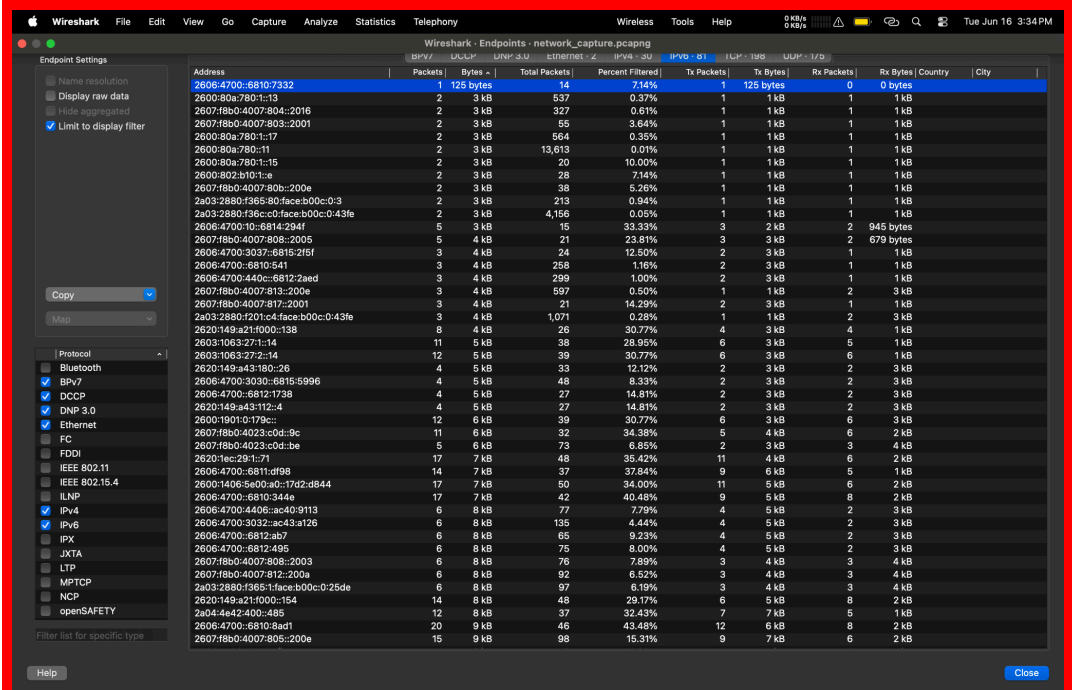
Figure 5. The protocol hierarchy showed the web browsing traffic relied heavily on TCP and TLS. TLS was expected since modern websites use HTTPS to encrypt communication

Endpoint Analysis

Wireshark's Endpoints feature was used to identify the devices and servers that participated in communication during the capture. The endpoint table displayed IP addresses, packet counts, and bytes transferred.

The local host communicated with multiple external servers during browsing activity. The endpoint with the highest packet count or byte count likely represented a frequently used web service, content delivery network, or video-streaming server.

Figure 6. Network endpoints in communication during the capture session.

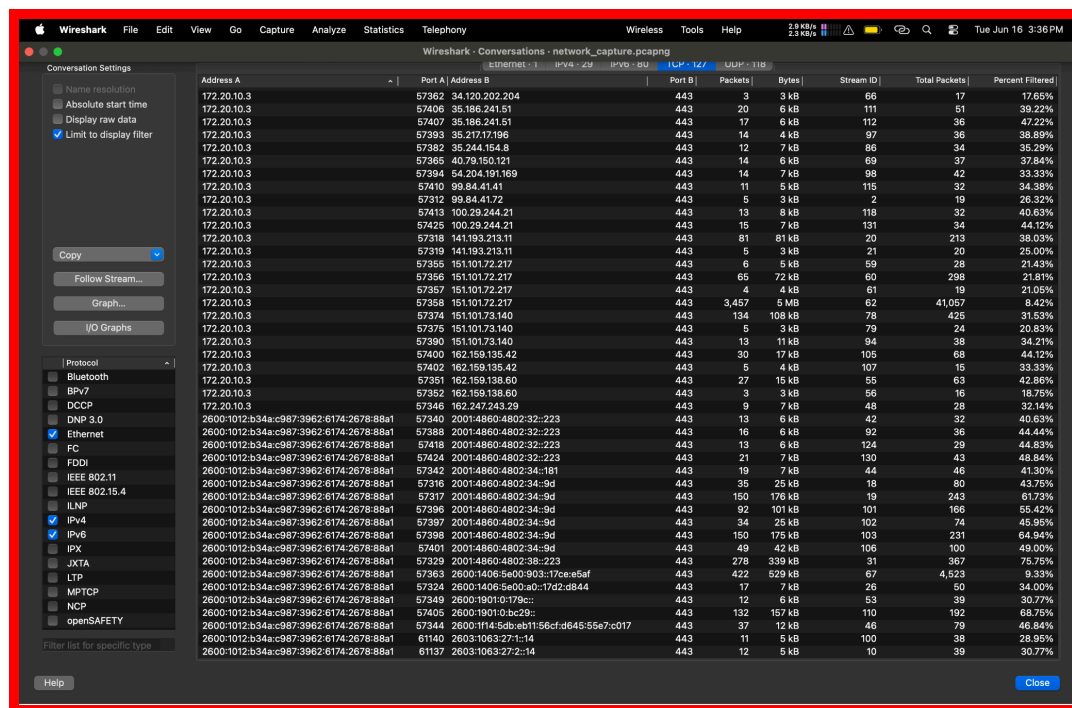


Address	Packets	Bytes	Total Packets	Percent Filtered	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City
2606:4700:6810:7332	1	125 bytes	14	7.14%	1	125 bytes	0	0 bytes		
2600:80a:780:1:13	2	3 kB	537	0.37%	1	1 kB	1	1 kB		
2607:f8b0:4007:804::2016	2	3 kB	327	0.61%	1	1 kB	1	1 kB		
2607:f8b0:4007:803::2001	2	3 kB	55	3.64%	1	1 kB	1	1 kB		
2600:80a:780:1:17	2	3 kB	564	0.35%	1	1 kB	1	1 kB		
2600:80a:780:1:11	2	3 kB	13,613	0.01%	1	1 kB	1	1 kB		
2600:80a:780:1:15	2	3 kB	20	10.00%	1	1 kB	1	1 kB		
2600:802:b10:1:e	2	3 kB	28	7.14%	1	1 kB	1	1 kB		
2607:f8b0:4007:80b::200e	2	3 kB	38	5.26%	1	1 kB	1	1 kB		
2a03:2880:f66:80:face:b00c:0:3	2	3 kB	213	0.94%	1	1 kB	1	1 kB		
2a03:2880:f66:c0:face:b00c:0:43fe	2	3 kB	4,156	0.05%	1	1 kB	1	1 kB		
2606:4700:10:6814:294f	5	3 kB	15	33.33%	3	2 kB	2	945 bytes		
2607:f8b0:4007:80b::2005	5	4 kB	21	23.81%	3	3 kB	2	679 bytes		
2606:4700:3037:6815:215f	3	4 kB	24	12.50%	2	3 kB	1	1 kB		
2606:4700:6810:541	3	4 kB	258	1.16%	2	3 kB	1	1 kB		
2606:4700:440c:6812:2aed	3	4 kB	299	1.00%	2	3 kB	1	1 kB		
2607:f8b0:4007:813::200e	3	4 kB	597	0.50%	1	1 kB	2	3 kB		
2607:f8b0:4007:817::2001	3	4 kB	21	14.29%	2	3 kB	1	1 kB		
2a03:2880:f201:c4:face:b00c:0:43fe	3	4 kB	1,071	0.28%	1	1 kB	2	3 kB		
2620:149:a2:1000:0:158	8	4 kB	26	30.77%	4	3 kB	4	1 kB		
2603:1063:27:1:14	11	5 kB	38	28.95%	6	3 kB	5	1 kB		
2603:1063:27:2:14	12	5 kB	39	30.77%	6	3 kB	6	1 kB		
2620:149:a43:180::26	4	5 kB	33	12.12%	2	3 kB	2	3 kB		
2606:4700:3030:6815:5996	4	5 kB	48	8.33%	2	3 kB	2	3 kB		
2606:4700:6812:1738	4	5 kB	27	14.81%	2	3 kB	2	3 kB		
2620:149:a43:112::4	4	5 kB	27	14.81%	2	3 kB	2	3 kB		
2600:1901:0:179c::	12	6 kB	39	30.77%	6	3 kB	6	3 kB		
2607:f8b0:4023:c0d:9c	11	6 kB	32	34.38%	5	4 kB	6	2 kB		
2607:f8b0:4023:c0d:be	5	6 kB	73	6.85%	2	3 kB	3	4 kB		
2620:1ec:29:1::71	17	7 kB	48	35.42%	11	4 kB	6	2 kB		
2606:4700:6811:d99	14	7 kB	37	37.84%	9	6 kB	5	1 kB		
2600:1406:5a00:a0:17d2:d844	17	7 kB	50	34.00%	11	5 kB	6	2 kB		
2606:4700:6810:344e	17	7 kB	42	40.48%	9	5 kB	8	2 kB		
2606:4700:4406:a040:9113	6	8 kB	77	7.79%	4	5 kB	2	3 kB		
2606:4700:3032:a043:a126	6	8 kB	135	4.44%	4	5 kB	2	3 kB		
2606:4700:6812:a07	6	8 kB	65	9.23%	4	5 kB	2	3 kB		
2606:4700:6812:495	6	8 kB	75	8.00%	4	5 kB	2	3 kB		
2607:f8b0:4007:80b::2003	6	8 kB	76	7.89%	3	4 kB	3	4 kB		
2607:f8b0:4007:812::200a	6	8 kB	92	6.52%	3	4 kB	3	4 kB		
2a03:2880:f66:1:face:b00c:0:26de	6	8 kB	97	6.19%	3	4 kB	3	4 kB		
2620:149:a2:1000:0:154	14	8 kB	48	29.17%	6	5 kB	8	2 kB		
2a04:4e42:400:485	12	8 kB	37	32.43%	7	7 kB	5	1 kB		
2606:4700:6810:bad1	20	9 kB	46	43.48%	12	6 kB	8	2 kB		
2607:f8b0:4007:805::200e	15	9 kB	98	15.31%	9	7 kB	6	2 kB		

Conversation Analysis

Wireshark's Conversations feature was used to examine communication sessions between the local device and remote servers. This view showed source and destination addresses, packet counts, and the amount of data exchanged.

Figure 7. Network conversations between the local host and external servers.



The largest conversation observed was between 172.20.10.3 and 151.101.72.217 over HTTPS.

The restore server is associated with a content delivery network used to deliver web content. The communication likely resulted from normal web browsing activity during the capture.

Graph Analysis

The I/O graph feature was used to visualize network traffic over time. Traffic spikes were expected when websites loaded resources, video streamed, or browser tabs requested additional content.

Figure 8. Network Traffic volume over time during web browsing activity.



The largest traffic spike occurred around 105 seconds. This was likely caused by watching a Youtube video / loading a site / downloading content

Key Findings

- DNS queries were observed for domains accessed during the browsing session
- TCP three-way handshakes established reliable connections between the local host and remote servers
- TLS 1.3 traffic protected web browsing activity through encryption.
- Most web communication used TCP and TLS because modern websites rely on HTTPS
- Endpoint and connection analysis identified multiple external servers involved in normal web browsing activity
- Network traffic increased during resource - heavy activity, such as loading websites or streaming video content.

Security Relevance

- Packet capture and network traffic monitoring
- DNS investigation
- TCP connection analysis
- TLS and HTTPS encryption analysis
- Identification of network endpoints
- Traffic-volume analysis
- Basic incident investigation documentation